Seah Kim

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Education

University of California, Berkeley 2019 – 2025
Ph.D. in Electrical Engineering and Computer Sciences
Dissertation: Scalable Specialization for Domain-Specific SoCs

Seoul National University 2014 – 2019
BS in Electrical and Computer Engineering

Research Interests

Computer Architecture, VLSI, SoC Design, Design Methodology, Systems for ML

Publications

MAVERIC: A Heterogeneous Robotics SoC with 4 CPU Cores and 13 INT8/FP32 Accelerators in 16-nm-Class Technology
Seah Kim, Jerry Zhao, Roger Hsiao, Yufeng Chi, Vighnesh Iyer, Vikram Jain, Borivoje Nikolić, Yakun Sophia Shao
IEEE Journal of Solid-State Circuits **(JSSC)**, April 2026.

MAVERIC: A 16nm 72 FPS, 10 mJ/frame Heterogeneous Robotics SoC with 4 Cores and 13 INT8/FP32 Accelerators
Seah Kim, Jerry Zhao, Roger Hsiao, Yufeng Chi, Vighnesh Iyer, Vikram Jain, Borivoje Nikolić, Yakun Sophia Shao
International Symposium on VLSI Technology and Circuits **(VLSI)**, June 2025.
Best Student Paper Award Finalist

SuperNoVA: Algorithm-Hardware Co-Design for Resource-Aware SLAM
Seah Kim, Roger Hsiao, Borivoje Nikolić, James Demmel, Yakun Sophia Shao
International Conference on Architectural Support for Programming Languages and Operating Systems **(ASPLOS)**, Volume 1, April 2025.
Artifact Review Badges: Artifact Available, Artifact Evaluated, Artifact Reproduced

AuRORA: A Full-Stack Solution for Scalable and Virtualized Accelerator Integration
Seah Kim, Jerry Zhao, Krste Asanović, Borivoje Nikolić, Yakun Sophia Shao
IEEE Micro **(Top Picks of 2023)**, July-August 2024.

AuRORA: Virtualized Accelerator Orchestration for Multi-Tenant Workloads
Seah Kim, Jerry Zhao, Krste Asanović, Borivoje Nikolić, Yakun Sophia Shao
International Symposium on Microarchitecture **(MICRO)**, October 2023.
ACM Artifact Review Badges: Artifact Available, Artifact Evaluated, Artifact Reproduced
Selected as one of "Top Picks from Computer Architecture Conferences, 2023"

RoSÉ: A Hardware-Software Co-Simulation Infrastructure Enabling Pre-Silicon Full-Stack Robotics SoC Evaluation
Dima Nikiforov, Shengjun Chris Dong, Chengyi Lux Zhang, **Seah Kim**, Borivoje Nikolić, Yakun Sophia Shao
International Symposium on Computer Architecture **(ISCA)**, June 2023.
ACM Artifact Review Badges: Artifact Available, Artifact Evaluated, Artifact Reproduced
ISCA Distinguished Artifact Award

DREAM: A Dynamic Scheduler for Dynamic Real-time Multi-model ML Workloads
Seah Kim, Hyoukjun Kwon, Jinook Song, Jihyuck Jo, Yu-Hsin Chen, Liangzhen Lai, Vikas Chandra
International Conference on Architectural Support for Programming Language and Operating Systems **(ASPLOS)**, Volume 4, April 2023.

MoCA: Memory-Centric, Adaptive Execution for Multi-Tenant Deep Neural Networks
Seah Kim, Hasan Genc, Vadim Vadimovich Nikiforov, Krste Asanović, Borivoje Nikolić, Yakun Sophia Shao
International Symposium on High-Performance Computer Architecture **(HPCA)**, March 2023.
IEEE Artifact Review Badges: Open Research Objects, Research Objects Reviewed, Results Reproduced

Gemmini: Enabling Systematic Deep-Learning Architecture Evaluation via Full-Stack Integration
Hasan Genc, **Seah Kim**, Alon Amid, Ameer Haj-Ali, Vighnesh Iyer, Pranav Prakash, Jerry Zhao, Daniel Grubb, Harrison Liew, Howard Mao, Albert Ou, Colin Schmidt, Samuel Steffl, John Wright, Ion Stoica, Jonathan Ragan-Kelley, Krste Asanović, Borivoje Nikolić, Yakun Sophia Shao
Design Automation Conference **(DAC)**, December 2021.
DAC Best Paper Award

Tutorials

Full-System, Full-Stack ML SoC Architecture Research with FireSim, Chipyard, Gemmini and AuRORA
Seah Kim, Abraham Gonzalez, Jerry Zhao, Joonho Whangbo, Vikram Jain
Full Day Tutorial at International Symposium on Microarchitecture (MICRO), November 2024.

Gemmini: Generate Custom DNN Accelerators with Full-System Full-Stack Evaluation
Hasan Genc, Simon Guo, **Seah Kim**, Vadim Nikiforov
Half Day Tutorial at Machine Learning and Systems (MLSys), August 2022.

Workshops, Invited Talks

Algorithm-Hardware Co-Design of SLAM with SuperNoVA
Seah Kim, Roger Hsiao, Borivoje Nikolić, James Demmel, Yakun Sophia Shao
3rd Workshop on Robotics Acceleration with Computing Hardware (RoboARCH) in conjunction with MICRO 2024.

Democratizing DNN Accelerators
Seah Kim
3rd Workshop on Democratizing Domain-Specific Accelerators (WDDSA) in conjunction with MICRO 2024.

AuRORA: Virtualized Accelerator Orchestration for Multi-Tenant Workloads
Seah Kim, Jerry Zhao, Krste Asanović, Borivoje Nikolić, Yakun Sophia Shao
9th Career Workshop for Inclusion and Diversity in Computer Architecture (CWIDCA) in conjunction with MICRO 2023.

An Open-source Framework for Virtualized and Disaggregated RISC-V Accelerators
Jerry Zhao, **Seah Kim**, Borivoje Nikolić, Krste Asanović, Yakun Sophia Shao
2nd Open-Source Computer Architecture Research (OSCAR) in conjunction with ISCA 2023.

Memory-Centric, Adaptive Execution for Multi-Tenant DNNs
Seah Kim, Hasan Genc, Vadim Vadimovich Nikiforov, Krste Asanović, Borivoje Nikolić, Yakun Sophia Shao
2nd Architecture, Compiler, and System Support for Multi-model DNN Workloads (ACSMD) in conjunction with ISCA 2022.

Gemmini: An Open-Source, Full-System DNN Accelerator Design and Evaluation Platform
Hasan Genc, **Seah Kim**, Vadim Vadimovich Nikiforov, Simon Zirui Guo, Borivoje Nikolić, Krste Asanović and Yakun Sophia Shao
1st Open-Source Computer Architecture Research (OSCAR) in conjunction with ISCA 2022.

Awards

VLSI Best Student Paper Award Finalist	2025
Sevin Rosen Funds Award	2025
MICRO PhD Forum	2024
Rising Stars in EECS	2024
Qualcomm Innovation Fellowship Finalist	2024
IEEE Micro's Top Pick in Computer Architecture	2024
Machine Learning and Systems Rising Star	2023
Qualcomm Innovation Fellowship Finalist	2023
ISCA Distinguished Artifact Award	2023
DAC Best Paper Award	2021
AI Compute Symposium Top Poster Award - IBM and IEEE CAS/EDS	2020
EECS Departmental Fellowship	2019

Study Abroad Fellowship - Kwanjeong Educational Foundation 2019-2023
National Scholarship for Science and Engineering - Korean Student Aid Foundation 2016-2018

Experience

[Industry]

Visiting Researcher, Meta, Menlo Park, CA Aug 2025 -

- AI and systems co-design team

AI Research Intern, Meta, Sunnyvale, CA May 2022 - Aug 2022

- Worked on creating the benchmark and dynamic scheduler for the AR/VR application
- Paper accepted to ASPLOS 2023

Special Project Group (SPG) Intern, Apple, Cupertino, CA May 2021 - Aug 2021

- Created a cycle-accurate performance model for sparse graph workload
- Designed an accelerator architecture for sparse graph workload

[Academic]

Graduate Student Researcher, UC Berkeley, Berkeley Architecture Research Group Nov 2019 - Aug 2025

- Advisors: Borivoje Nikolić, Yakun Sophia Shao

Graduate Student Researcher, UC Berkeley, Berkeley Wireless Research Center Aug 2019 - Aug 2025

- Led an Intel16 chip tape-out (first-author): design submitted in December 2023, successfully tested in Summer 2024

Undergraduate Researcher, Integrated Systems Design Lab, Seoul National University Jan 2018 - July 2019

- Advisor: Deog-Kyoon Jeong
- Participated in a Samsung 28nm and a TSMC 40nm CMOS tape-out of a fractional digital phase locked loop using Injection Locking Oscillators
- Designed a transmitter for an automotive imaging sensor

Teaching Experiences

Guest Lecturer for Hardware for Machine Learning (EE 290) Spring 2024
Graduate Student Instructor for Digital Design and Integrated Circuits (EECS 151/251A) Fall 2022
Graduate Student Instructor for Great Ideas in Computer Architecture (CS 61C) Fall 2020
Tutor for Analog Integrated Circuits (Seoul National University) Fall 2018

Service

HPCA Program Committee 2026
UC Berkeley KGSA x KSEA undergraduate student mentoring 2024
IISWC Artifact Evaluation Committee 2024
Visit Day Area Student Lead Organizer 2024
Visit Day Peer Advisor 2021-2024
Visit Day Area Student Organizer 2021-2023
Campus Mentoring Program (Seoul National University) 2015
Class Representative (Department of ECE, Seoul National University) 2014

Skills

Languages: C++, C, Scala, Chisel, Verilog, SystemVerilog, Python, RISC-V Assembly, Make, Bash, Pytorch, TensorFlow

Hadware Platforms, Tools: Cadence Physical Design Tools (Genus, Innovus, Virtuoso, Joules), Vivado, Verdi, VCS, Xilinx FPGAs, Arduino, Raspberry Pi